

AWAN ABADULLAH

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PROFILE

A self-driven and versatile **Systems Engineer** with experience delivering complex automotive and robotics systems from concept through validation and release. Proven track record at **Stellantis** in system-level design, validation, and cross-functional integration across North American vehicle brands including **RAM**, **DODGE**, and **JEEP**.

Recognized as a **system-level SME** and **knowledge resource**, with experience guiding new hires, supporting architecture/process transitions, and translating complex system behavior into clear requirements, risks, and execution plans for cross-functional teams.

Comfortable navigating evolving requirements, defining interfaces, coordinating with suppliers, and aligning technical work with timing, quality, and performance targets.

Skilled in **IBM DOORS**, **MS Visio**, **SharePoint**, **Python**, and **structured documentation**, with a strong command of traceability, compliance tracking, system risk evaluation, and issue resolution.

DFSS **Green Belt** certified in **Systems Behavior Testing**, with additional experience in embedded systems, rapid prototyping, and humanoid robotics, bringing a **multidisciplinary problem-solving mindset** to engineering teams.

SKILLS

Systems Engineering: Systems Architecture Design, Interface Control, Requirements Management, Systems Behavior Testing, Failure Mode Analysis, CAN interfacing, Compliance Matrix Generation, Risk Assessment

Technical Leadership: Cross-functional coordination, technical training, new-hire onboarding, SME support, root cause analysis, recurring issue resolution, risk mitigation, stakeholder alignment

Tools and Software: IBM DOORS, GitHub, Issue Tracking Systems (JIRA, RTC), MS Visio, SharePoint, SolidWorks, AutoCAD, OnShape, MATLAB, EAGLE, MS Office, MS PowerApps, Copilot

Mechanical and Electrical Integration: Embedded Systems, Hardware-Software Interfacing, Electro-mechanical System Validation

Programming Languages: Python, VBA, C++, JavaScript, MATLAB, HTML/CSS

Operating Systems: Windows, Mac, Linux

WORK EXPERIENCE

Systems Engineer

April 2022 - Present

Stellantis Canada, Ontario, Canada (Remote)

- Managed system design and integration for 13+ vehicle features across Stellantis North America platforms, using IBM DOORS, MS Visio, SharePoint, and structured requirements documentation.
- Recognized as an SME for active chassis braking functions, regularly supporting engineers, technical leads, and management with system behavior clarification, process guidance, and issue resolution.
- Trained 15+ engineers on system processes, documentation standards, feature workflows, and DOORS/Visio practices; supported new-hire onboarding and technical handover.
- Led development of Off-Road Cornering Assist and supported High Rail Mode feature development, coordinating Braking, HMI, Powertrain, Controls, validation, and supplier stakeholders from concept through release readiness.
- Prototyped and pitched an AI-based DFMEA assistant through Stellantis' Bright Idea initiative, leading software system design and implementation to extract failure modes from specifications and target ~60% reduction in DFMEA preparation time.
- Led documentation cleanup and standardization across system artifacts, coordinating multiple teams to improve requirements accuracy, traceability, and reuse.

- Resolved a late-stage ESC button logic issue across 12+ trim levels by replacing flawed PROXI logic with LIN-based control in coordination with EE leads and validation chiefs.
- Led resolution of 10+ high-priority GIMS issues by organizing stakeholder reviews, tracking technical risk, and driving closure through structured root cause analysis.

Robotics Systems Engineering Technician

Sept 2021 - Jan 2022

Sanctuary AI, Vancouver, Canada (Hybrid)

- Integrated humanoid robotic subsystems including hands, heads, torsos, wrist assemblies, and embedded components across mechanical, electrical, and software interfaces.
- Built and tested prototype stress rigs for humanoid wrist and arm motion, supporting failure identification, design iteration, and early validation activities.
- Collaborated with electrical engineers, mechanical designers, robotic technicians, and suppliers to troubleshoot subsystem behavior and support full-platform integration.
- Developed Python-based automation tools to summarize technical research and reduce manual R&D review effort.
- Authored Confluence documentation on hydraulics, thumb kinematics, component behavior, and integration procedures to support knowledge transfer and future development.

EDUCATION

Bachelor's in Mechatronics Systems Engineer (Dean's List)

Sept 2017 - April 2021

Western University, Ontario, Canada

PROJECTS

Advanced Off-Road Feature Integration, Stellantis

- Led system design and integration for an off-road driving feature across braking, HMI, and powertrain modules.
- Authored requirements and interface documentation in IBM DOORS and MS Visio; supported CAN analysis and DFSS-based validation.

"MAPSY" Mobile Autonomous Pool Cleaner, Western University

- Designed a four-in-one robotic system for pool maintenance, including surface skimming, vacuuming, and automated chemical disinfection.
- Programmed sensor-based propulsion in C/C++ and built a MATLAB image pipeline for dirt detection.
- Received the Innovative Design Prize at the Western Engineering Competition for functional integration and systems complexity.

LED Chest Plate Indicator System, Sanctuary AI

- Designed and integrated an embedded LED state-indicator system for a humanoid robot prototype.
- Supported hardware/software integration, component selection, and sourcing for robot state communication.
- The robot featuring this system was later featured in Bloomberg as part of early media coverage on humanoid robotics innovation.

CERTIFICATIONS & COURSEWORK

- **Design for Six Sigma (DFSS): Systems Behavior Testing Green Belt** (via Stellantis)
- **Powertrain Electrification (Level 1), Power Electronics and Machines (Level 2)** – Ohio State University (via Stellantis)
- **Certified SolidWorks Associate (CSWA and CSWA-S)** in Mechanical Design & Simulation
- **The Power of Communication** – Vin Giang's Stage Academy